

# Reading List for the Algebraic Islands Project

Prioritized by relevance to current research

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This reading list supports the paper “The Algebraic Landscape of Kochen–Specker Sets in Dimension Three” and its companion PRL letters. Papers are ordered by priority within four tiers. Current results to keep in mind while reading: OCUS proof that 31 is optimal in the 49-ray integer pool; universal merge saturation across all six islands (3,756 merges);  $N(S) = 6$  for the Eisenstein island (confirming the Cortez–Morales–Reyes minimal ring); SI-C closure convergence with the integer pool.

## Tier 1 — Read Carefully

These papers directly shape the arguments and claims in our work.

1. **S. Trandafir and A. Cabello**, “Two fundamental solutions to the rigid Kochen–Specker set problem and the solution to the minimal Kochen–Specker set problem under one assumption,” *Phys. Rev. A* **111**, 052204 (2025). arXiv:2501.11640.

*Why:* The single most relevant paper. Proves CK-31 is rigid (unique up to unitary equivalence). Constructs the 97-element SI-C closure—our computation shows it overlaps our integer pool in 37 rays and contains it after a second closure round. Introduces bootstrap percolation for KS sets (Proposition 1). Their Conjecture 1 (no rigid KS set  $\leq 30$  containing the SI-C set) is complementary to our OCUS proof. Re-read with the SI-C closure and  $N(S)$  results fresh.

2. **V. Cortez, R. Morales, and E. J. Reyes**, “Minimal ring extensions of the integers exhibiting Kochen–Specker contextuality,” arXiv:2211.13216 (2022).

*Why:* Proves  $Z[1/6]$  is the minimal ring extension of  $Z$  exhibiting KS contextuality in dimension 3. Their  $N(S) = \text{lcm}\{\|v\|^2 : v \in S\}$  invariant is the bridge between their ring-theoretic framework and our algebraic islands: our computation confirms the Eisenstein island achieves exactly  $N = 6$ , the minimal value. Understanding their proof technique (Theorem 1:  $6 \mid N$  is necessary and sufficient) could help prove the  $6 \mid n$  conjecture for cyclotomic fields.

3. **Z. Li, C. Bright, and V. Ganesh**, “Leveraging SAT solvers for the minimum Kochen–Specker problem,” *Proceedings of IJCAI-24*, pp. 1944–1952 (2024). arXiv:2306.13319.

*Why:* Establishes the current lower bound of 24 vectors via SAT+CAS orderly generation. Their pipeline (enumerate abstract KS hypergraphs  $\rightarrow$  check realizability via Z3) is directly complementary to our finite-pool SAT encoding (C4a–C4c), which could serve as a “quick

accept” screen. Key details: 17 known minimal unembeddable subgraphs, 41 KS candidates at  $n = 23$  (all unembeddable), estimated  $\sim 125$  CPU-years to push to  $n = 24$ . Read carefully if pursuing the LBG filter avenue.

4. **M. Kirchweger, A. Peitl, and S. Szeider**, “Co-certificate learning with SAT modulo symmetries,” *Proceedings of AAAI-23*, pp. 4119–4126 (2023).

*Why*: Independent lower bound of 24. Co-certificate learning (CCL) is conceptually related to our OCUS/CEGAR blocking—both learn structural constraints from satisfying assignments. Key data: 3,747,950 KS candidates at  $n = 23$ , of which only 2 survive their unembeddable-subgraph filter. The explosive growth (88,282 at  $n = 22 \rightarrow 3.7\text{M}$  at  $n = 23$ ) shows why pure enumeration cannot reach  $n = 31$ .

## Tier 2 — Read for Depth

These strengthen connections already made in our papers.

5. **A. Cabello**, “Simplest bipartite perfect quantum strategies,” *Phys. Rev. Lett.* **134**, 010201 (2025). arXiv:2311.17735.

*Why*: Definition 2 (B-KS) is the foundation of our BPQS analysis. Theorem 1 (every BPQS defines a KS set) gives operational significance to all six islands. A careful re-read ensures our encoding of conditions (I') and (II') is airtight.

6. **A. Cabello**, “Simplest Kochen–Specker set,” *Phys. Rev. Lett.* (2025). arXiv:2508.07335.

*Why*: The 33-vector, 14-basis Eisenstein construction via Weyl–Heisenberg. Table I has BPQS numbers we independently verify. This is the concrete bridge between our Eisenstein island and Cabello’s “simplest” criterion. His coordinates  $\{0, 1, \omega, \bar{\omega}\}$  are precisely our Eisenstein alphabet.

7. **Y. Liu, H. Y. Chung, E. Z. Cruzeiro, J. R. Gonzales-Ureta, R. Ramanathan, and A. Cabello**, “Equivalence between face nonsignaling correlations, full nonlocality, all-versus-nothing proofs, and pseudotelepathy,” *Phys. Rev. Research* **6**, L042035 (2024). arXiv:2310.10600.

*Why*: The chain  $\text{FNS} = \text{FN} = \text{AVN} = \text{PT}$ , all equivalent to KS sets. This is what makes our island classification operationally significant: each island defines not just a KS set but an entire class of perfect quantum strategies.

## Tier 3 — Read for New Avenues

These open new directions of research.

8. **S. Abramsky, R. S. Barbosa, and S. Mansfield**, “The contextual fraction as a measure of contextuality,” *Phys. Rev. Lett.* **119**, 050504 (2017). arXiv:1705.07918.

*Why*: The contextual fraction is a different measure from CSW—it quantifies the “amount” of contextuality via a linear program. Computing it for each island would give a third axis (alongside CSW and BPQS) for the Pareto frontier across algebraic substrates.

9. **C. Budroni, A. Cabello, O. Gühne, M. Kleinmann, and J.-Å. Larsson**, “Kochen–Specker contextuality,” *Rev. Mod. Phys.* **94**, 045007 (2022). arXiv:2102.13036.

*Why:* The comprehensive review ( $\sim 100$  pages). Essential for ensuring we haven’t missed a connection and for framing the introduction and related work. Particularly relevant: Section IV (dimension-specific results), Section VI (operational significance), and the open problems list.

10. **A. Peres**, *Quantum Theory: Concepts and Methods* (Kluwer, 1993), Chapter 7.

*Why:* The original source for CK-31, with cube diagrams showing how the coordinate alphabet  $\{0, \pm 1, \pm 2\}$  generates the construction. Worth revisiting to ensure our alphabet-based narrative is faithful to the original presentation.

## Tier 4 — Skim for Awareness

Background and historical context.

11. **M. Pavicic, N. D. Megill, B. C. Waegell, and P. K. Aravind**, “Automated generation of Kochen–Specker sets,” *Sci. Rep.* **9**, 6765 (2019).

*Why:* Their automated generation approach and “vector component” terminology (our “coordinate alphabet”). Relevant mainly to avoid the terminological dispute (Pavicic distinguishes “KS sets” from “extended KS sets”; we use standard definitions).

12. **S. Uijlen and B. Westerbaan**, “A Kochen–Specker system has at least 22 vectors,” *New Generation Computing* **34**, 3–23 (2016).

*Why:* Historical context only—the previous lower bound (22), now superseded by the 24 of Li–Bright–Ganesh and Kirchweger et al.

## Reading Strategy

- Papers 1–4 deserve careful, line-by-line reading. Budget 2–3 hours each.
- Papers 5–7 are quick re-reads to verify specific claims. Budget 30–60 minutes each.
- Papers 8–10 are for whichever new avenue you choose to pursue.
- Papers 11–12 are 15-minute skims.

*Total estimated reading time: 15–20 hours.*